Check your understanding

1. It is important to stir the water regularly when heating the water. Explain why. [1 mark]

   .....

2. A student repeats the experiment for a block of copper four times.

   The results for the temperature of the block after 10 minutes are: 87 °C, 71 °C, 68 °C and 69 °C.

   a. Calculate the mean maximum temperature. Take into account any anomalous results. [2 marks]

   .....

   .....

   b. Give one reason why the student may have anomalous results. [1 mark]

   .....

3. If the heating element is not in contact with the metal block, there will be a layer of air in between the heating element and the metal block. The calculated specific heat capacity may then be **greater** than the actual value.

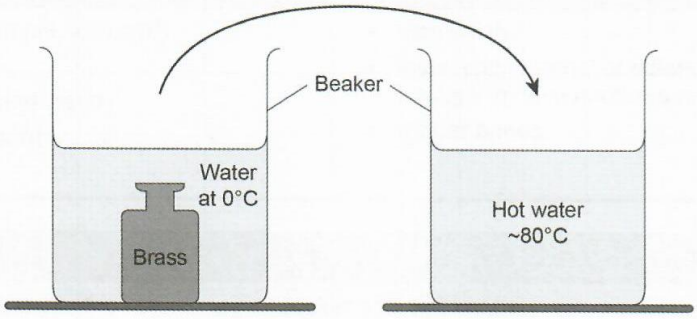
   Explain why. [1 mark]

   .....

4. Another method to calculate specific heat capacity of an object is to immerse the object in ice cold water until the object reaches 0 °C and then to place it in a beaker of hotter water.

   In **Figure 2**, a brass block is moved from water at 0 °C to hot water at 80 °C.

   Figure 2



   List three pieces of equipment that would be needed to calculate the specific heat capacity of the brass. [3 marks]

   .....

   .....

Exam-style questions

1. 1 kg of aluminium was heated for 10 minutes.
Table 3 shows the results.

Table 3

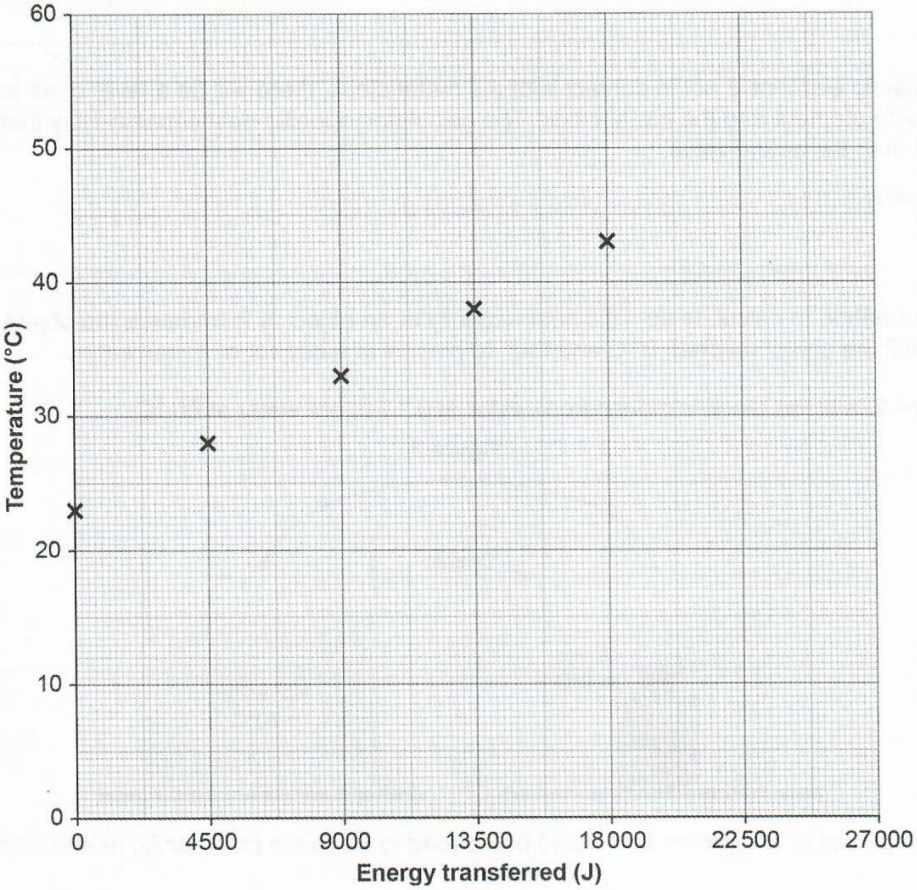
Energy transferred (J)	Temperature (°C)
0	23
4500	28
9000	33
13 500	38
18 000	43
22 500	48
27 000	53

a. Complete **Graph 2** using data from **Table 3**.

- Plot the additional data points.
- Draw a line of best fit.

[3 marks]

Graph 2



b. Determine the specific heat capacity of this block of aluminium. Use the gradient of the line and the formula below.

$$\text{specific heat capacity} = \frac{1}{\text{mass} \times \text{gradient}}$$

[2 marks]

.....

.....

2. The specific heat capacity for three different metals is listed in **Table 4**.

Table 4

Metal	Specific heat capacity (J/kg °C)
Aluminium	897
Copper	385
Iron	450

All three materials are commonly used as cooking pans. Which material would heat up the fastest? Give a reason for your answer. [2 marks]

.....

.....

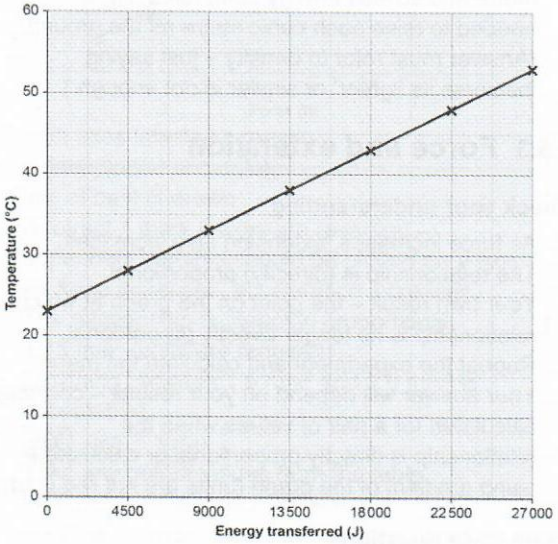
4.1.1.3 Specific heat capacity

Check your understanding

1. to ensure that the entire mass of the water is heated evenly [1]
2. **a.** 69 °C (69.3 °C also allowed) [2]
   b. Any one from: [1]
   - misreading of thermometer
   - fault with equipment
3. The thermal energy from the heating element may be transferred to the surrounding air rather than to the block of metal. [1]
4. balance (accept scale) [1]
   measuring cylinder [1]
   thermometer [1]

Exam-style questions

1. **a.** Your graph should look something like this:



   1 mark for plotting 1 or 2 correct points [1]
   1 mark for all 3 points correctly plotted [1]
   1 mark for line of best fit [1]

   b. gradient of line = 30/27 000 = 0.0011 [1]
   (allow a mark for any other set of values that give a gradient of 0.0011)
   specific heat capacity = $1/(1 \times 0.0011)$
   = 909.1 [1]
2. Copper would heat up the fastest. [1]
   As it has the lowest specific heat capacity it needs less energy to raise 1 kg by 1 °C. [1]